

Numerical Results

31 March 2026

1 Experimental Setting

$$S_0 = 100, \quad v_0 = 0.114, \quad v'_0 = 0.110, \quad r = 0, \quad T = 1,$$

with

$$\kappa_1 = 5.5, \quad \kappa_2 = 0.1, \quad \theta = 0.078, \quad \xi_1 = 2.689, \quad \xi_2 = 0.502,$$

and

$$\delta_1 = \delta_2 = 0.94, \quad \rho_{12} = -0.982, \quad \rho_{13} = -0.727, \quad \rho_{23} = 0.59.$$

The Bermudan contract is sampled at 12 exercise dates. Five strike levels are considered:

$$K \in \{70, 80, 90, 100, 110\},$$

corresponding to deeper OTM through ITM puts.

All comparisons in this note use the fixed benchmark references generated with 1200 Euler steps. The reference direct prices are

$$\begin{aligned} V_{110}^{\text{ref}} &= 16.645784, & V_{100}^{\text{ref}} &= 11.356823, & V_{90}^{\text{ref}} &= 7.402999, \\ V_{80}^{\text{ref}} &= 4.596899, & V_{70}^{\text{ref}} &= 2.699339. \end{aligned}$$

For any direct price estimate V and reference value V^{ref} , the direct relative error is defined by

$$\varepsilon_{\text{dir}} = \frac{|V - V^{\text{ref}}|}{|V^{\text{ref}}|}.$$

The reported confidence-interval width is $2 \times 1.96 \times \text{SE}$.

The purpose of the study is to compare plain LSMC with the Hybrid LSMC–PDE method under *matched path budgets*. Two matched-path regimes are examined:

- 60,000 direct paths for both methods;
- 20,000 direct paths for both methods.

The Euler step counts are 24, 48, 72, 96. The emphasis is on the direct estimator, because this is where the hybrid method displayed its clearest advantage.

2 Matched 60,000-Path Comparison

Table 1 summarizes the step counts at which the Hybrid LSMC–PDE method achieves a smaller direct error than plain LSMC when both methods use 60,000 direct paths.

The main feature is that the hybrid advantage is strongest in the middle of the step grid, especially around 48 to 72 steps.

Table 2 reports representative values at 72 steps. This is the most informative point because it lies in the strongest favorable region while avoiding the edge effects of 24 and 96 steps.

The direct-error picture is favorable to the hybrid method in the middle of the step grid. The variance picture is even clearer: for every strike in Table 2, the hybrid method has a substantially smaller standard error and a much narrower confidence interval, even though its runtime remains far larger than that of plain LSMC.

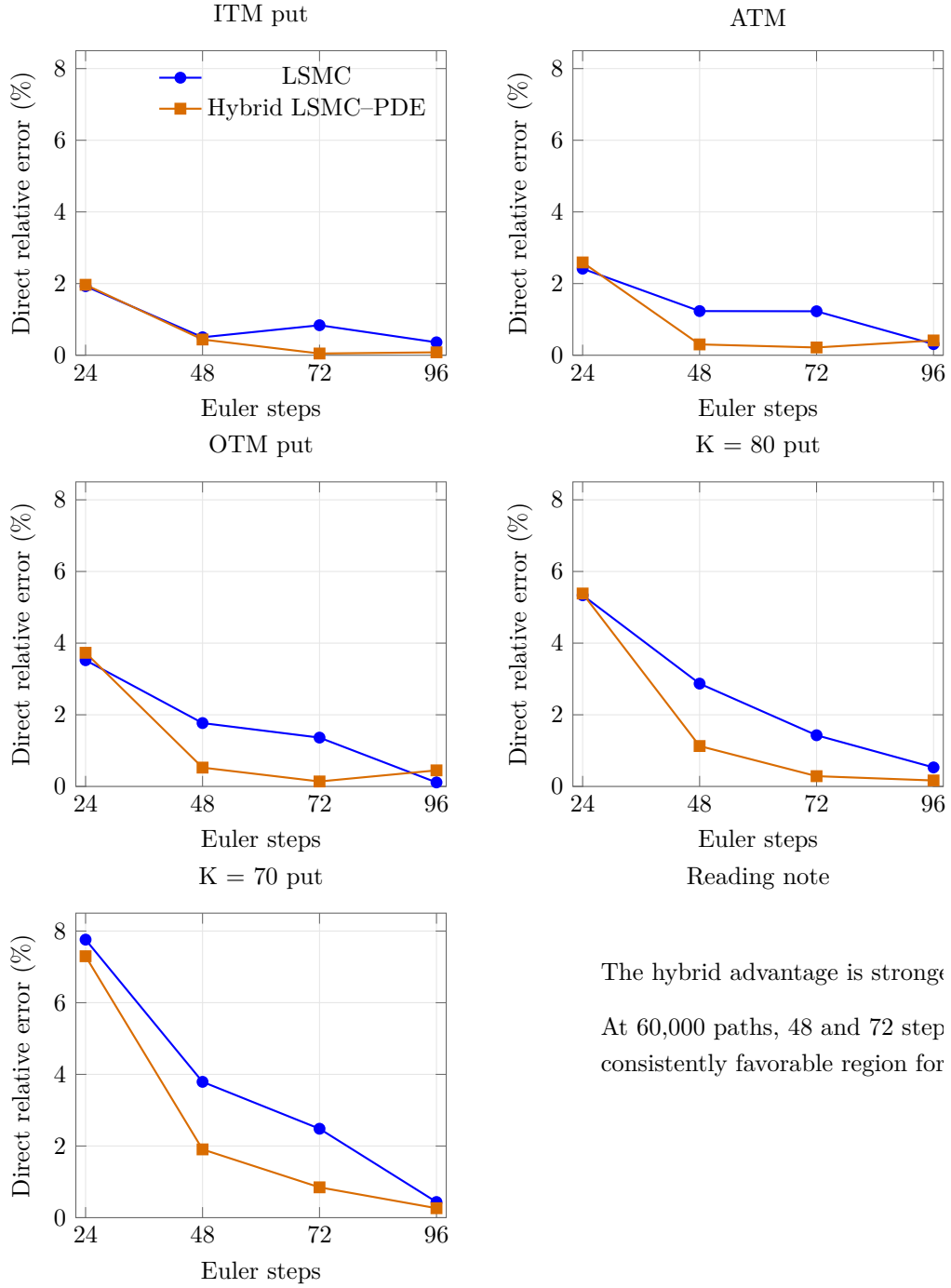


Figure 1: Direct relative error under the matched 60,000-path comparison.

Table 1: Steps at which the hybrid method is more accurate under the matched 60,000-path comparison.

Scenario	Hybrid better at steps
ITM put	48, 72, 96
ATM	48, 72
OTM put	48, 72
K = 80 put	48, 72, 96
K = 70 put	24, 48, 72, 96

3 Matched 20,000-Path Comparison

The matched path budget was then reduced from 60,000 to 20,000 paths for both methods. This is a more demanding test because both methods receive less Monte Carlo information.

Reducing the path budget strengthens the favorable direct-error result. The benchmark LSMC method deteriorates faster than the hybrid method, so the hybrid advantage becomes broader across the step grid.

At 20,000 paths, the same qualitative conclusion persists and becomes stronger: the hybrid method usually attains lower direct error with much tighter confidence intervals, while the runtime gap remains significant.

4 Overall Conclusions

- The matched-path comparison at 60,000 paths shows that the Hybrid LSMC–PDE method is most favorable in the middle of the Euler grid, especially at 48 and 72 steps.
- The matched-path comparison at 20,000 paths strengthens this result. When the path budget is reduced, the benchmark LSMC method deteriorates faster than the hybrid method, so the hybrid direct-error advantage becomes broader.
- In both matched-path regimes, the hybrid method has much smaller standard errors and much narrower confidence intervals than plain LSMC across the full strike range.
- The main trade-off remains runtime: the hybrid method is much slower than LSMC, even though it is often more accurate and more stable under equal path budgets.

Taken together, the experiments carried out on 31 March 2026 support a clear message: when the comparison is made on a fair equal-path basis, the Hybrid LSMC–PDE method performs particularly well at moderate Euler resolutions, and its variance advantage becomes even more valuable when the path budget is tightened.

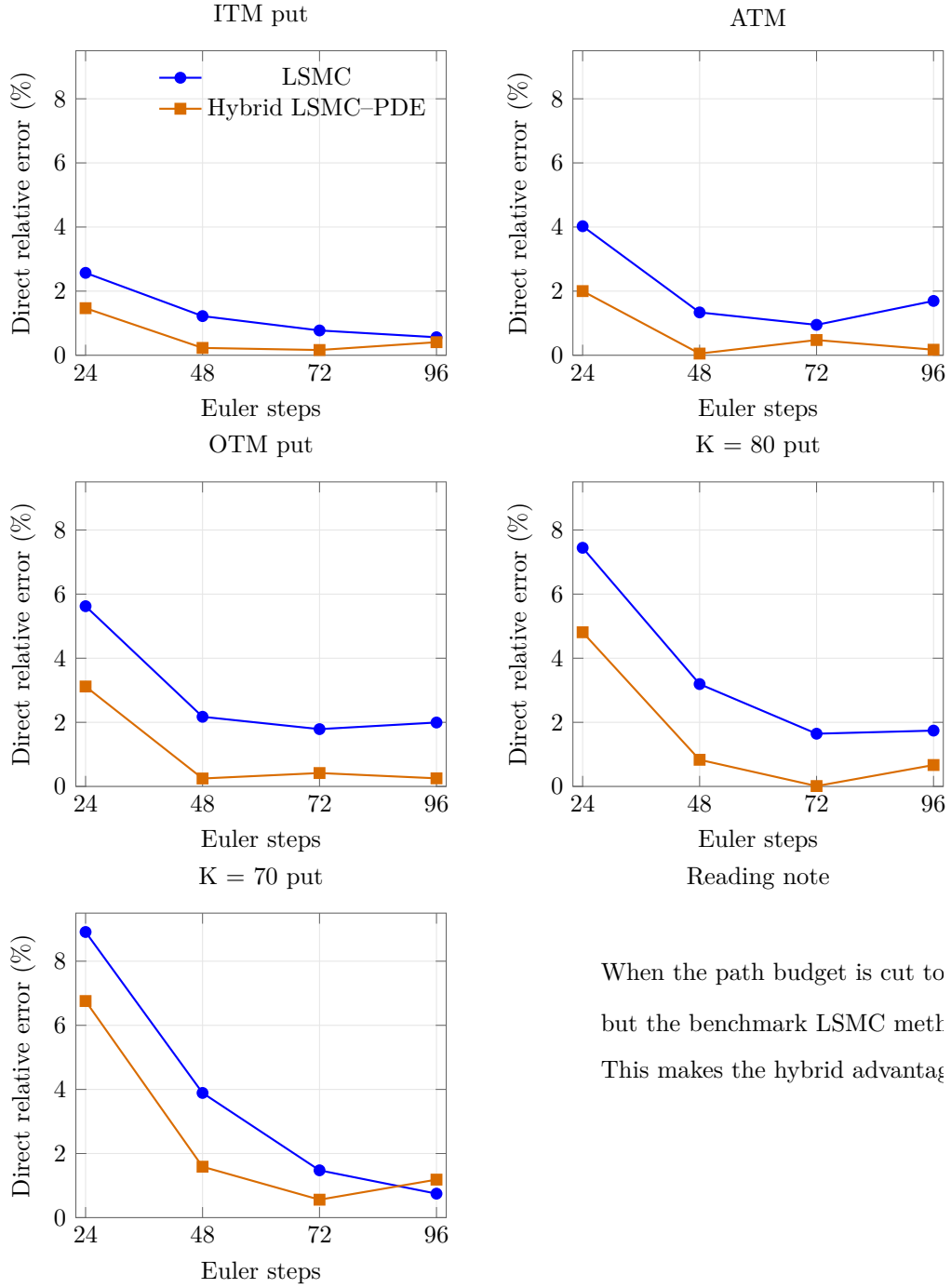


Figure 2: Direct relative error under the matched 20,000-path comparison.

Table 2: Representative metrics at 72 Euler steps under the matched 60,000-path comparison.

Scenario	LSMC				Hybrid LSMC–PDE			
	Dir. err.	SE	CI width	Runtime	Dir. err.	SE	CI width	Runtime
ITM put	0.837%	0.080321	0.314858	1.18 s	0.048%	0.026208	0.102736	238.75 s
ATM	1.227%	0.069157	0.271095	1.25 s	0.216%	0.020668	0.081018	241.84 s
OTM put	1.363%	0.056750	0.222460	1.16 s	0.139%	0.015294	0.059951	240.15 s
K = 80 put	1.429%	0.044924	0.176102	1.17 s	0.287%	0.010629	0.041668	230.58 s
K = 70 put	2.482%	0.034333	0.134585	1.21 s	0.847%	0.006908	0.027079	233.90 s

Table 3: Steps at which the hybrid method is more accurate under the matched 20,000-path comparison.

Scenario	Hybrid better at steps
ITM put	24, 48, 72, 96
ATM	24, 48, 72, 96
OTM put	24, 48, 72, 96
K = 80 put	24, 48, 72, 96
K = 70 put	24, 48, 72

Table 4: Representative metrics at 72 Euler steps under the matched 20,000-path comparison.

Scenario	LSMC				Hybrid LSMC–PDE			
	Dir. err.	SE	CI width	Runtime	Dir. err.	SE	CI width	Runtime
ITM put	0.773%	0.134824	0.528510	0.43 s	0.163%	0.045393	0.177942	76.22 s
ATM	0.948%	0.117430	0.460326	0.46 s	0.475%	0.035909	0.140765	78.05 s
OTM put	1.788%	0.098666	0.386771	0.50 s	0.418%	0.026674	0.104563	78.74 s
K = 80 put	1.645%	0.076738	0.300813	0.49 s	0.012%	0.018645	0.073089	78.68 s
K = 70 put	1.476%	0.057802	0.226584	0.50 s	0.560%	0.012206	0.047847	78.67 s